Staff answers for tectonic hazards

Tuesday 23 February 2027 | P2 | 40 minutes | Year A

Humanities | LAUNCH | Spring 2 Week 1 | HUM-LAUNCH-S2-W01

Core explanation

C1: divergent plates move apart. Earthquakes arise from brittle cracking or fault slip. Rising mantle undergoes decompression melting. C2: at a subduction zone fault locking and release produce earthquakes; slab-derived water lowers the melting temperature of mantle rock above the slab, helping produce magma. C3: transform motion can lock and slip, producing earthquakes, with no usual magma-forming mechanism. Do not teach the slab simply “melts through friction”.

Japan example

C4: 11 March 2011, Tohoku region, Japan. Pacific plate subduction beneath northern Japan, a locked interface, continuing convergence and accumulated strain, then sudden fault slip and waves. Current USGS event catalogue magnitude is 9.1. Some older primary summaries report 9.0. Accept either with its stated source; use 9.1 consistently in this pack. The upper plate is described at different model scales, so its name is not required here.

Route answers

Supported: an ordered earthquake chain, a separate subduction-volcano chain, and the Japan date and mechanism with appropriate actual support. Independent: add a correct transform comparison. Aspire: divergent melting follows reduced pressure whereas subduction melting is aided by water; some earthquakes occur within plates, hot spots can create volcanoes away from boundaries, and continental collision does not imply a volcanic arc. All routes explain both hazards and the real event. An unsupported label or copied chain does not show an independent explanation.

Checks

Hinge: At a transform boundary the plates slide past. Why can an earthquake happen? A: Hot weather melts the plate. B: Rock can lock, strain builds, then sudden slip releases waves. C: Every earthquake is caused by a volcano. Answer B.

Parallel check: The ocean plate keeps moving slowly, but part of a subduction fault is stuck. What builds before it slips? Answer: strain or stress in the rocks. Sudden slip releases energy as waves. Ask each learner to finish the because link without the model.

Evidence and sensitive response

Offer to the Communication portfolio. Keep the learner’s own idea, an actual adult audience reply, one NS and the learner’s response in the same colour, on the same page or usual mode before R. E follows actual evidence banking. Record actual help. A partner’s answer cannot establish both learners’ understanding.

Use calm diagrams and precise language. No disaster footage, casualty stories, family prompts or hazard simulations. Acknowledge distress without probing. Follow the planned supervised pause and normal safeguarding route to the safeguarding lead if a concern is disclosed.

Prompt ladder: 1 Point to place/instruction. 2 Ask short question. 3 Rephrase instruction. 4 Provide scaffold/choice. 5 Model first step only, then return task.

Actual observation _____ Help _____
Gap and planned follow-up _____ Actual banking _____